SG No. 202 T_{h}^{3} $Fm\bar{3}$ [cubic]

* plus set: $+[0,\,0,\,0]$, $+\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$

Table 1: Wyckoff site: `4a`, site symmetry: `m-3.`

No.	position	mapping
1	$[0,\,0,\,0]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `4b`, site symmetry: `m-3.`

No.	position	mapping
1	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`

Table 3: Wyckoff site: `8c`, site symmetry: `23.`

No.	position	mapping
1	$\left[\frac{1}{4},\,\frac{1}{4},\,\frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{3}{4},\,\frac{3}{4},\,\frac{3}{4}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 4: Wyckoff site: `24d`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[0,\,\frac{1}{4},\,\frac{1}{4}\right]$	`[1,4,13,16]`
2	$\left[0,\,\frac{3}{4},\,\frac{1}{4}\right]$	`[2,3,14,15]`
3	$\left[\frac{1}{4},\,0,\,\frac{1}{4}\right]$	`[5,8,17,20]`
4	$\left[\frac{1}{4},\,0,\,\frac{3}{4}\right]$	`[6,7,18,19]`
5	$\left[\frac{1}{4},\,\frac{1}{4},\,0\right]$	`[9,12,21,24]`
6	$\left[\frac{3}{4},\,\frac{1}{4},\,0\right]$	`[10,11,22,23]`

Table 5: Wyckoff site: `24e`, site symmetry: `mm2..`

No.	position	mapping
1	$[x,\,0,\,0]$	`[1,4,14,15]`
2	$[-x,\,0,\,0]$	`[2,3,13,16]`
3	$[0,\,x,\,0]$	`[5,8,18,19]`
4	$[0,\,-x,\,0]$	`[6,7,17,20]`
5	$[0,\,0,\,x]$	`[9,12,22,23]`
6	$[0,\,0,\,-x]$	`[10,11,21,24]`

Table 6: Wyckoff site: `32f`, site symmetry: `.3.`

No.	position	mapping
1	$[x,\,x,\,x]$	`[1,5,9]`
2	$[-x,\,-x,\,x]$	`[2,7,12]`

continued ...

Table 6

No.	position	mapping
3	$[-x,\ x,\ -x]$	`[3,8,10]`
4	$[x,\ -x,\ -x]$	`[4,6,11]`
5	$[-x,\ -x,\ -x]$	`[13,17,21]`
6	$[x,\ x,\ -x]$	`[14,19,24]`
7	$[x,\ -x,\ x]$	`[15,20,22]`
8	$[-x,\ x,\ x]$	`[16,18,23]`

Table 7: Wyckoff site: `48g`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,4]`
2	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[2,3]`
3	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	`[5,8]`
4	$\left[\frac{1}{4},\ -x,\ \frac{3}{4}\right]$	`[6,7]`
5	$\left[\frac{1}{4},\ \frac{1}{4},\ x\right]$	`[9,12]`
6	$\left[\frac{3}{4},\ \frac{1}{4},\ -x\right]$	`[10,11]`
7	$\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[13,16]`
8	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[14,15]`
9	$\left[\frac{3}{4},\ -x,\ \frac{3}{4}\right]$	`[17,20]`
10	$\left[\frac{3}{4},\ x,\ \frac{1}{4}\right]$	`[18,19]`
11	$\left[\frac{3}{4},\ \frac{3}{4},\ -x\right]$	`[21,24]`
12	$\left[\frac{1}{4},\ \frac{3}{4},\ x\right]$	`[22,23]`

Table 8: Wyckoff site: `48h`, site symmetry: `m..`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,16]`
2	$[0,\ -y,\ z]$	`[2,15]`
3	$[0,\ y,\ -z]$	`[3,14]`
4	$[0,\ -y,\ -z]$	`[4,13]`
5	$[z,\ 0,\ y]$	`[5,20]`
6	$[z,\ 0,\ -y]$	`[6,19]`
7	$[-z,\ 0,\ y]$	`[7,18]`
8	$[-z,\ 0,\ -y]$	`[8,17]`
9	$[y,\ z,\ 0]$	`[9,24]`
10	$[-y,\ z,\ 0]$	`[10,23]`
11	$[y,\ -z,\ 0]$	`[11,22]`
12	$[-y,\ -z,\ 0]$	`[12,21]`

Table 9: Wyckoff site: `96i`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`

continued ...

Table 9

No.	position	mapping
2	$[-x, -y, z]$	[2]
3	$[-x, y, -z]$	[3]
4	$[x, -y, -z]$	[4]
5	$[z, x, y]$	[5]
6	$[z, -x, -y]$	[6]
7	$[-z, -x, y]$	[7]
8	$[-z, x, -y]$	[8]
9	$[y, z, x]$	[9]
10	$[-y, z, -x]$	[10]
11	$[y, -z, -x]$	[11]
12	$[-y, -z, x]$	[12]
13	$[-x, -y, -z]$	[13]
14	$[x, y, -z]$	[14]
15	$[x, -y, z]$	[15]
16	$[-x, y, z]$	[16]
17	$[-z, -x, -y]$	[17]
18	$[-z, x, y]$	[18]
19	$[z, x, -y]$	[19]
20	$[z, -x, y]$	[20]
21	$[-y, -z, -x]$	[21]
22	$[y, -z, x]$	[22]
23	$[-y, z, x]$	[23]
24	$[y, z, -x]$	[24]